What a Margin of Error Does and Does Not Cover

Everything that goes wrong outside the sampling

84-355 Democracy's Data

August 2026

5 states in 2024, carrying 64 electoral votes between them, were decided by a margin narrower than the **±3.10 points** a standard poll of a thousand people advertises as its own precision. The electoral college that year was 312 to 226; those 64 votes were more than enough to reverse it.

The ± attached to a poll is a specific, narrow, mathematically exact promise. Almost nobody who reads it knows what it promises, and the races that decide American elections sit inside it.

01 Polls are not merely reported, they are acted on

Polls are not merely reported; **they are acted on.** They decide which candidates are treated as viable, which states get campaign money, which get visits, and which arguments are made about whether an election was normal. A poll error is not a private embarrassment for a pollster — it redistributes attention and resources before anyone votes.

And the errors are not evenly distributed. They matter most where the race is closest, which is precisely where the polls are least able to help.

02 Whether the polling industry is in crisis is genuinely disputed

The dispute is real. One reading is that three consecutive presidential cycles understated the same candidate, that response rates have fallen below 1%, and that the instrument has broken. The other is that national polling error in recent cycles has been within its historical range. On this reading, critics grade polls against a standard polls never claimed to meet, namely calling the winner, and the visible failures are concentrated in state polls with thin budgets.

The simulation below does not settle that. It establishes something narrower and harder to argue with: **the quantity the industry publishes as its uncertainty is not the quantity that has been going wrong.**

03 One real file, and a country that was manufactured on purpose

This chapter has two data objects and they could hardly be less alike. The first is real: 51 rows of 2024 presidential results by state, with each state's electoral votes and each candidate's share. It was copied from the electoral-map chapter's folder so that this one stands alone, and it traces back to a private compilation of the fifty-one separate state canvasses. As the election-returns chapter explains at length, no federal agency produces a national table of American election results. It carries percentages and electoral votes and no vote counts at all, which is enough for what it is used for here and would not be enough for much else.

The second object has no source, because **it was built.** The population polled below was stipulated: a million voters split at a fixed number, and then sampled. That is the design and not a compromise, and the reason is worth stating flatly.

You cannot watch a poll fail unless you already know the right answer, and with real polling data you never do. Not on the day the poll is published, and not afterwards either: by the time the election settles the question, the population has changed and the poll cannot be re-run against it. Holding the truth fixed and varying exactly one thing is the only way to separate the error a poll advertises from the error that has actually been going wrong. No archive of real polls can do that, however large.

Two honest disclosures follow, and the first is an exception to a rule this book otherwise keeps. **The stipulated split is typed, not computed.** It sits in the setup chunk as a constant, chosen to sit near the 2024 national two-party result. It could not be worked out from the file kept here, because that file holds percentages by state and no counts to add up.

The second is that **the response gap is invented**. Twenty percent is a plausible-looking number, not a measured one; the shape of what it does is what transfers, and the size of it transfers nowhere. The later sections say what else follows from those two choices.

What that buys is a kind of cleaning question that mostly does not arise: there was nothing to clean. The state file arrived tidy and was copied unchanged. The population was made here rather than collected, so it has no missing values, no coding conventions and nobody else's decisions inside it. That is precisely why it cannot stand in for a real electorate.

That is the price of making a dataset instead of finding one. **It can only be asked the questions its author already knew to build into it.** This population has three dials: how many people are interviewed, how differently the two sides answer the phone, and what the pollster reweights on. Those are the three things the chapter can demonstrate.

It has no dial for a likely-voter screen that guesses wrong, or a question worded to lead, or a sample that never reaches the people it should. It is silent on all of them, and its silence should not be mistaken for their absence.

The one precaution is that every draw in the chapter runs from a fixed random seed, so the figures are the same each time the page is built. The central result is checked against algebra rather than against a draw. A reader who wants the real version of this argument should go to the Cooperative Election Study. It publishes tens of thousands of interviews per cycle, with the weights attached and a written account of how they were built.

04 The file arrived clean because somebody else cleaned it

"There was nothing to clean" is true of the file kept here and false of the data. The cleaning happened one folder over, and it is worth looking at, because a file that arrives tidy is the most dangerous kind: there is no friction to make you ask who decided what.

What the compilation actually ships is not a CSV at all. It is `pres_by_state.rda`, an R binary object on GitHub. Loading it and asking R what is inside gives this:

Column	What it holds	Stored as	First values
year	the presidential election year	number	1864 1864 ...
state_abbrev	the state's two-letter code	text	"CA" "CT" ...
winner	who carried the state	text	"Abraham Lincoln" "Abraham Lincoln" ...
party_win	that winner's party	text	"republican" "republican" ...
democrat	the Democratic share of the vote, as a percentage	number	41.4 48.6 ...
other	the share going to everyone else	number	NA NA NA NA NA ...
republican	the Republican share	number	58.6 51.4 ...

1,913 rows, 7 columns, every presidential election from 1864 to 2024 in one object. Here are four of the 2024 rows:

Row	Year	State abbrev	Winner	Party win	Democrat	Other	Republican
1863	2024	AK	Donald Trump	republican	41.41	1.68	54.54
1864	2024	AL	Donald Trump	republican	34.10	NA	64.57
1865	2024	AR	Donald Trump	republican	33.56	1.12	64.20
1866	2024	AZ	Donald Trump	republican	46.70	0.50	52.20

Now the same four states in `pres2024_states.csv`, the file this chapter opens:

State	Abbrev	Ev	Harris	Trump	Other	Winner	Margin
Alabama	AL	9	34.10	64.57	0.00	Trump	30.47
Alaska	AK	3	41.41	54.54	1.68	Trump	13.13
Arizona	AZ	11	46.70	52.20	0.50	Trump	5.50
Arkansas	AR	6	33.56	64.20	1.12	Trump	30.64

Set the two side by side and every difference is a decision somebody made.

The unit changed. 1,913 state-elections became 51 states, by keeping one election and discarding every other one in the file. Nothing warns you that the discarded rows exist; the clean file has no `year` column at all, so a reader cannot tell from it which election it describes.

A party became a candidate. The raw columns are `democrat` and `republican` — *parties*, correctly, because a table that runs back to Lincoln cannot have a `harris` column. The build renames them `harris` and `trump`, which is right for 2024 and would be nonsense for any other row in the source. That rename is the whole reason the file cannot be extended without being rebuilt.

A blank became a zero. Alabama's `other` is `NA` in the source and `0` here. Those are not the same claim: one says nobody voted for anyone else, the other says the compiler did not record it. The build chose the first reading, and because `other` is never summed in this chapter the choice does no damage — but it is a choice, and it is invisible in the clean file.

Two columns were typed in by hand. Electoral votes are not in the source and are not derivable from it; they are the 2020 apportionment, keyed in as a constant and checked against 538. The `col` and `row` columns place each state on the tile grid a few pages down, and they were invented outright. They are an opinion about where Delaware goes, and no data anywhere implies them.

And `margin` is arithmetic, `trump - harris` rounded to two places, computed because the question needs it and no source publishes it.

What none of that produces is vote counts. The source carries percentages, so the clean file carries percentages, and no amount of care downstream can put back a number that was never in the file. That is why the stipulated split at the heart of this chapter had to be typed rather than derived: **there is nothing here to sum.**

05 The margin a poll would have had to resolve

Start with the real data, because it sets the scale of what a poll is being asked to do.

State	Code	Electoral votes	Harris (%)	Trump (%)	Other (%)	Winner	Margin (Trump – Harris)
Pennsylvania	PA	19	48.66	50.37	0	Trump	1.71

One row per state, 51 rows, 538 electoral votes. The margin column is the thing a poll is trying to estimate. In Pennsylvania it was 1.71 points.

Here is every state whose actual result came in tighter than the ±3.10 a standard national poll of 1,000 people advertises.

State	Electoral votes	Harris (%)	Trump (%)	Margin (Trump – Harris)	Winner
Wisconsin	10	48.74	49.60	0.86	Trump
Michigan	15	48.31	49.73	1.42	Trump
Pennsylvania	19	48.66	50.37	1.71	Trump
Georgia	16	48.53	50.73	2.20	Trump
New Hampshire	4	50.65	47.87	-2.78	Harris

The tightest of them, Wisconsin, came in at 0.86 points — a tenth of the interval a poll would have brought to the question.

Here is the whole country arranged as a grid: one square per state, roughly where the state sits on the map. No square is bigger than any other, so Wyoming is as visible as California.

One square per state, shaded by how close it was. Black outline: decided by less than the ±3.10 a poll of 1,000 claims.

5 of the 51 squares are outlined, together 64 electoral votes. The tightest was Wisconsin at 0.86 points.

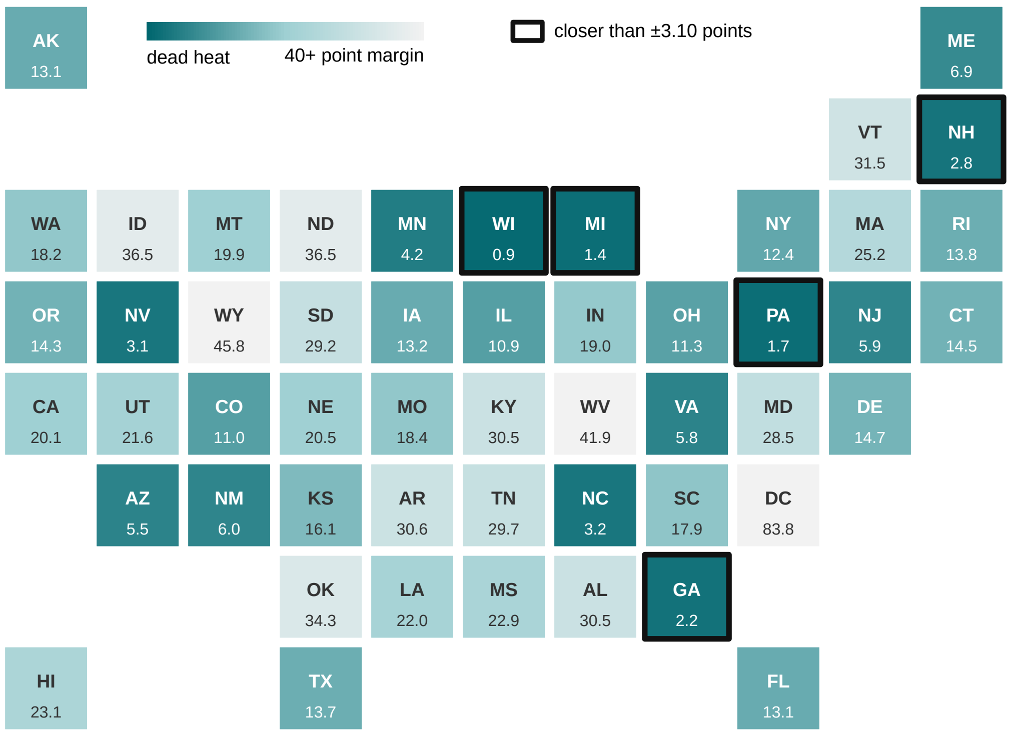


Figure 1. The 2024 presidential election, one square per state. Squares are placed roughly where the state sits on the map and shaded by how close the result was, so Wyoming is as visible as California. The 5 squares with a black outline were decided by less than the ±3.10 points a poll of 1,000 advertises; together they carry 64 electoral votes. The tightest was Wisconsin, at 0.86 points.

The races that decide American elections sit inside the error bar of the instrument used to describe them, which means everything about how that error bar is computed matters. Working out what it does and does not cover requires something no real election supplies: a case where the answer is known before the poll is taken.

06 A country we can see all of

So build one. Construct 1,000,000 voters with a **stipulated** split: Harris 49.25%, Trump 50.75%, roughly the 2024 two-party result. Then poll them.

Candidate	Voters	Share
Harris	492,500	49.25
Trump	507,500	50.75

That number is an input, not a finding. It is the one unrealistic thing here, and it is the only thing that makes the exercise possible: we know the answer, so we can grade every poll below.

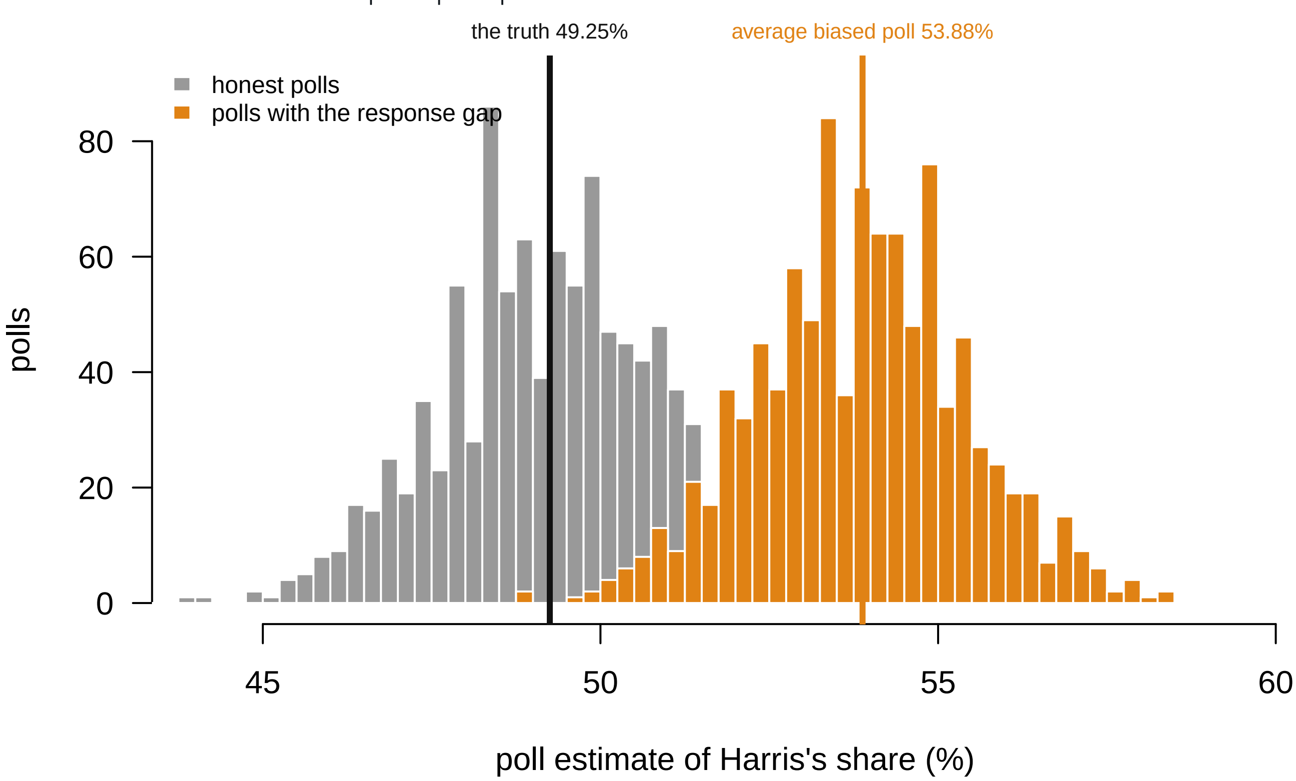
07 Three honest polls, three answers

Poll	Estimate of Harris's share (%)	Truth (%)
1	50.90	49.25
2	52.10	49.25
3	48.20	49.25

Three honest polls of 1,000 people, drawn perfectly at random, and three different answers. That wobble is **sampling error**, and it is the only kind of error the ± describes.

08 The promise, and whether it is kept

Run a thousand honest polls, and beside them a thousand polls drawn from a pool where Harris voters are 20% likelier to pick up the phone.



Gray: 1,000 honest polls of 1,000 people. Orange: the same 1,000 polls when Harris voters are 20% likelier to answer. Both are tight; only one is right.

Figure 2. A thousand honest polls, and the same thousand polls with a response gap. Each poll asks 1,000 people. Gray: honest polls, drawn perfectly at random, centered on

the truth at 49.25%. Orange: the same polls when Harris voters are 20% likelier to answer, centered on 53.88%. **Both distributions are equally tight.** Only one of them is in the right place, and nothing about the width of either says which.

Quantity	Value
Margin of error claimed at n = 1,000	±3.10 points
Polls landing inside it	94.5%
Average of the thousand polls	49.27%
The truth	49.25%

The promise is kept exactly. About 95% of the polls land inside ±3.10, and the average of them is the truth to two decimal places.

Now read the promise carefully, because it is narrower than it sounds. It says: *if my only problem is that I asked 1,000 people instead of all of them, I will usually be within three points.* **It says nothing whatever about any other problem.**

09 Bigger polls are more precise

People polled	Margin of error (points)
500	±4.38
1,000	±3.10
5,000	±1.39
20,000	±0.69
100,000	±0.31

Familiar, and true. Precision improves with the square root of the sample, so a hundredfold increase buys a tenfold improvement. At 100,000 respondents the poll advertises **±0.31 points** — a claim of near-certainty.

> **Before you read on, commit to a number.** Everything about that table assumed that when you call somebody, they answer. In 2024 the response rate to a typical telephone poll was **under 1%**. Suppose the 1% who pick up are not a random 1% — suppose Harris voters are 20% likelier to answer.
>
> **How far off will a poll of 100,000 people be?** Write down a number of points, in the same units as the ±0.31 it advertises, before you read on.

10 The same poll, when people decline to answer

People polled	Estimate (%)	Advertised margin	Actually wrong by	As a multiple of its own margin
1,000	54.00	±3.10	+4.75	1.5×
10,000	54.63	±0.98	+5.38	5.5×
100,000	53.68	±0.31	+4.43	14.3×

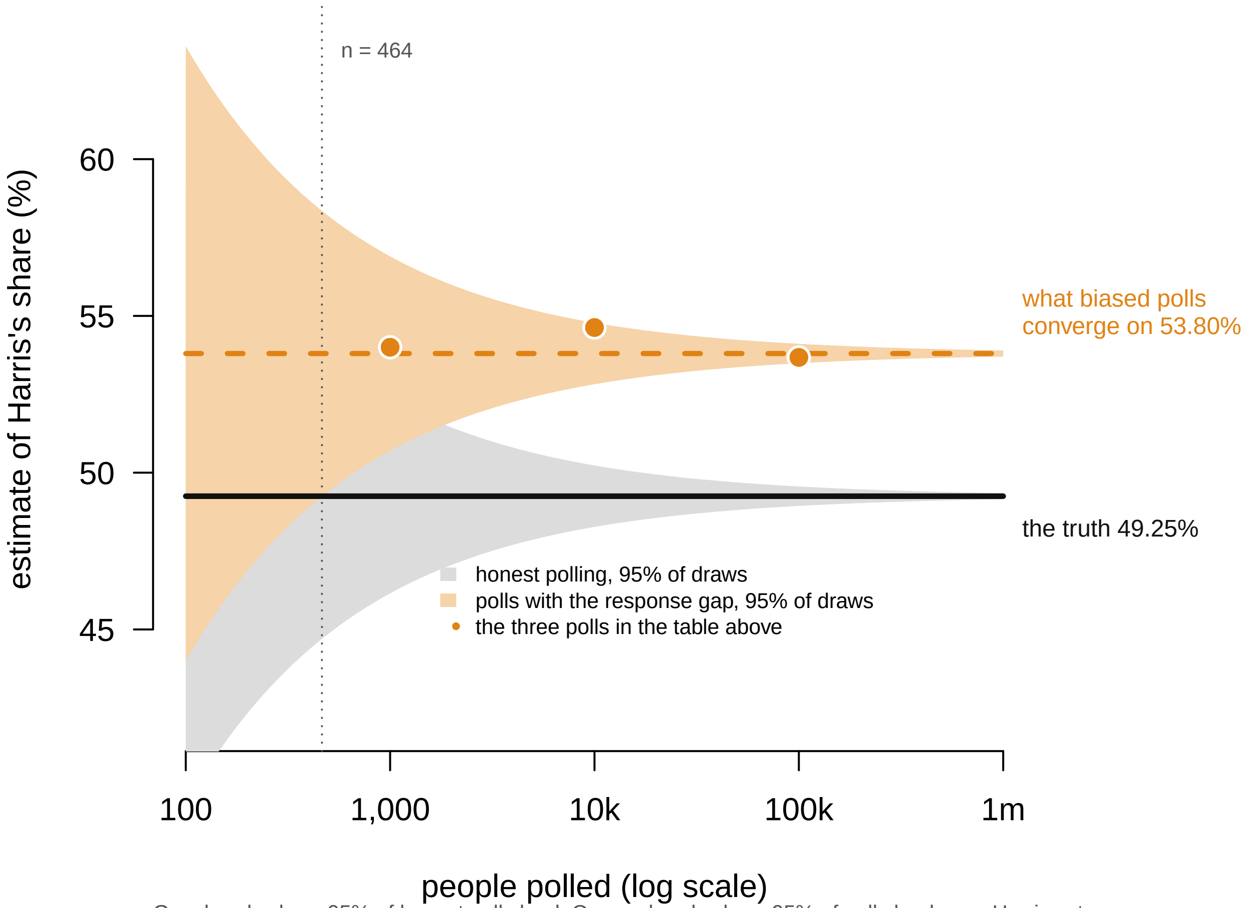
Most readers write down something smaller than the ±3 they started with, reasoning that a poll a hundred times larger cannot be worse than a small one. A margin of error is a claim about how far off a poll could be.

A poll of 100,000 people reports a margin of error of ±0.31 points and misses by 4.43 — about 14 times its own stated precision.

And it does not drift toward the truth as the sample grows. It **converges** — on 53.80%, which is not the answer.

Quantity	Value
The truth	49.25%
What the poll converges on as $n \to \infty$	53.80%
The bias	+4.55 points
Margin of error at $n = 100{,}000$	±0.31 points

Drawn as a funnel, the two behaviors are impossible to confuse. Each band is where 95% of polls of that size land.



Gray band: where 95% of honest polls land. Orange band: where 95% of polls land once Harris voters are 20% likelier to answer, and the orange dots are the three polls in the table above. Past about 464 respondents the orange band is narrower than the 4.55−point bias, so from there on the poll's own interval never contains the truth again, and it gets worse, not better, with every extra respondent.

Figure 3. Where polls land, as the sample grows. The horizontal axis is the number of people polled, on a log scale. The gray band is where 95% of honest polls fall. The orange band is where 95% fall once Harris voters are 20% likelier to answer, and the orange dots are the three polls in the table above.

Both bands narrow at the same rate, but only one of them narrows onto the truth. Past about 464 respondents the orange band is narrower than the 4.55-point bias. From there on **the poll's own interval never contains the truth again**, and it gets worse with every extra respondent.

That value is algebra, not a lucky draw: if Harris voters answer 20% more readily, the responding pool is 53.80% Harris no matter how many of them you talk to.

More data makes you more confident about the wrong number. The tight clustering in the orange distribution above is not evidence of accuracy. It is evidence of agreement, and every one of those polls agrees on a figure 4.55 points from the truth. Averaging them changes nothing, because they are all wrong in the same direction. That is exactly what a polling average does, and exactly why a polling average's narrow interval does not mean what readers take it to mean.

11 How small does the gap have to be?

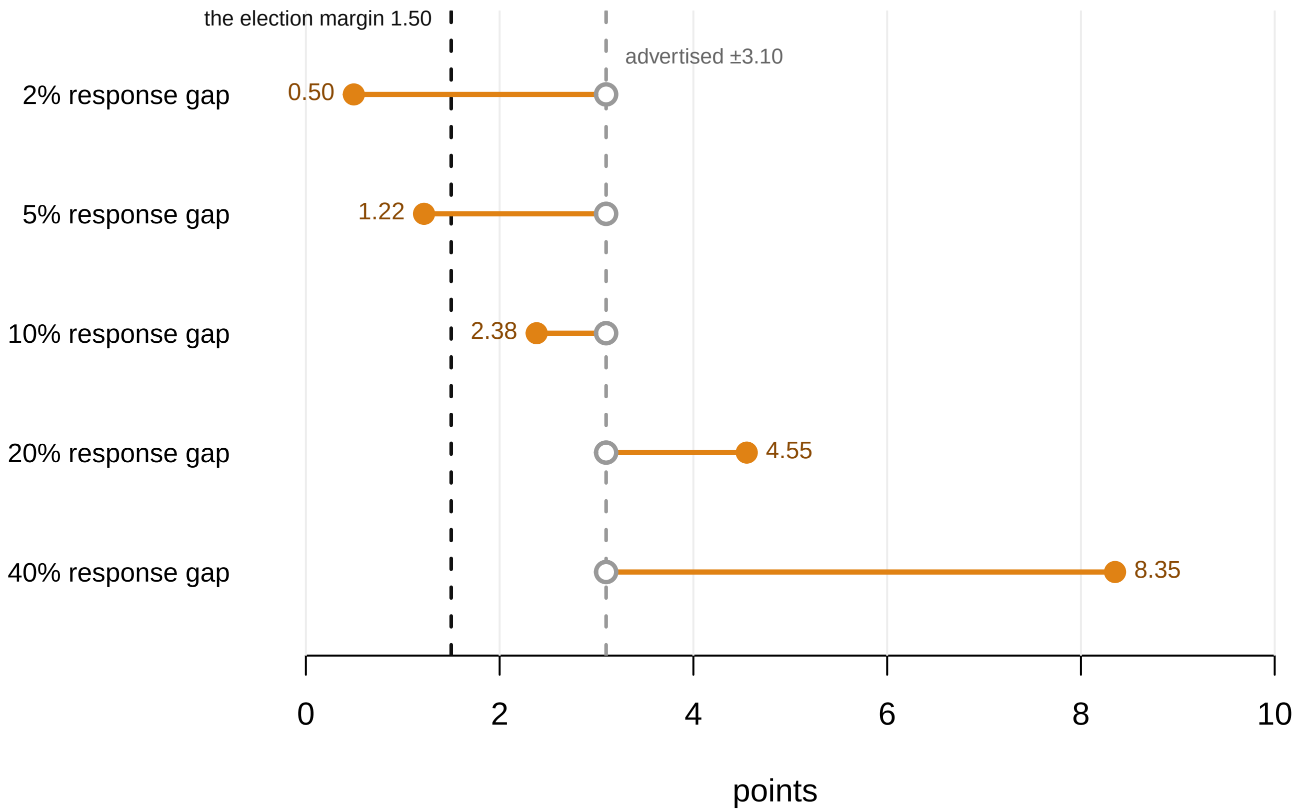
20% sounds like a large distortion. It is not obvious that it is.

Response gap	Bias (points)	Advertised margin at n = 1,000	As a multiple of the simulated election margin
2%	+0.50	±3.10	0.3
5%	+1.22	±3.10	0.8
10%	+2.38	±3.10	1.6
20%	+4.55	±3.10	3.0
40%	+8.35	±3.10	5.6

A 5% difference in willingness to answer the phone produces 1.22 points of bias. No pollster could detect a gap that size, and it sounds like nothing. The two-party margin this simulation is built around is 1.50 points.

The temptation is to look at that row and be reassured, because 1.22 is smaller than ±3.10. That is the wrong comparison. **The bias should be measured against the size of the thing you are trying to detect**, not against the size of a different error you have already accounted for. Against a 1.50-point election, it is most of the answer.

Note also that the two quantities behave differently. The advertised margin shrinks with sample size and does not respond to the gap at all; the bias responds to the gap and does not respond to sample size. **They are not commensurable, and publishing only one of them is a choice.**



Hollow gray dot: the ±3.10 a poll of 1,000 advertises, the same on every row, because sample size is all it responds to. Filled orange dot: the bias the response gap actually produces. 2 of the 5 gaps put the bias past the advertised margin, and 3 of 5 put it past the 1.50−point margin of the election itself.

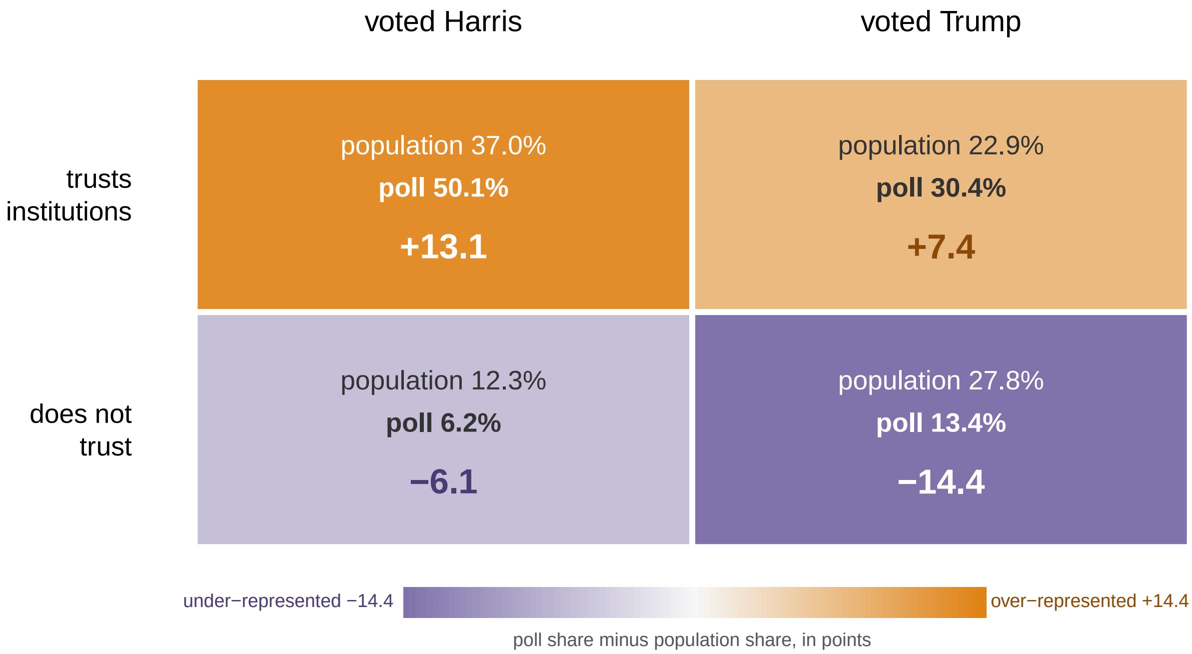
Figure 4. What each response gap costs, against what the poll advertises. One row per response gap. The hollow gray dot is the ±3.10 a poll of 1,000 advertises — identical on every row, because sample size is the only thing it responds to. The filled orange dot is the bias that gap actually produces. 2 of the 5 gaps put the bias past the advertised margin, and 3 of 5 put it past the 1.50-point margin of the election itself.

12 Weighting works when you already know what is wrong

Pollsters know all of this, and they correct for it. If you know one group is over-represented, count its answers for less. Build a population where response is driven by something other than the vote, say trust in institutions, and try.

Estimate	Harris share (%)	Error (points)
The truth	49.28	—
Raw poll	56.26	+6.98
Weighted on trust	49.98	+0.70
Weighted on a variable unrelated to response	56.26	+6.98

The mechanism is easiest to see as a four-cell grid. Two things are true of every simulated voter: how they voted, and whether they trust institutions. The poll reaches the four kinds of person at very different rates.



Poll share minus population share. Harris is 49.3% of the country and 56.3% of the raw poll.

Orange cells are over−represented in the poll, purple under−represented. Nothing in this grid is about the vote directly: the poll over−samples people who trust institutions, and it is only because trust and vote are correlated that Harris comes out at 56.3% of the poll against 49.3% of the country. Weight the two rows back to their true sizes and the error falls to +0.70 points.

Figure 5. Who the poll reached, and who it should have reached. One cell per combination of vote and trust in institutions, shaded by the poll's share of that cell minus the population's: orange where the poll over-represents, purple where it under-represents. Nothing in the grid is about the vote directly. The poll over-samples people who trust institutions. It is only because trust and vote go together that Harris comes out at 56.3% of the poll against 49.3% of the country.

Weighting means counting some answers more than others, to correct a sample that does not match the country. Weighting on the thing that actually drives non-response takes the error from +6.98 to +0.70. Weighting on a variable that is real, measurable and irrelevant moves the estimate essentially not at all — +6.98.

So weighting works, exactly and only when you already know what is wrong. After 2016 the industry discovered it had not been weighting by education and added it. In 2020 it missed again, in the same direction. The variable you have not thought of is invisible by construction, and no amount of arithmetic finds it.

13 The margin of error describes the smallest problem

A margin of error is a statement about sample size and nothing else. It is honest within its own terms: the coverage check above found 95% of honest polls inside it, exactly as promised. And it is silent about every other source of error a poll can have.

Put the two quantities side by side and the asymmetry is the whole point. With a 20% response gap, a poll of 100,000 people advertises ±0.31 and carries a bias of 4.55 points. No increase in sample size reduces the second number by anything at all, and every increase shrinks the first — which is why the two grow further apart the more money a pollster spends.

None of which licenses the reverse inference. Nothing here says real polls are biased by 4.55 points, or that Harris voters really do answer 20% more often. **Those were inputs, chosen to make the mechanism visible.** The shape of the argument transfers; the size of any particular number in it does not. A simulation is not evidence about the world.

14 The formal version, and the empirical case

Meng (2018) gives the formal version of the funnel above, and it is sharper than the simulation. Meng breaks survey error into three factors: data quality, data quantity, and problem difficulty. The factor measuring the link between responding and the thing being measured is multiplied by the square root of the *population* size. The consequence he calls the **big data paradox**: *the more the data, the surer we fool ourselves.* On that calculation, a self-reported 1% sample of American voters in 2016, about 2.3 million people, carried the same mean squared error as a genuine random sample of roughly 400.

Bradley et al. (2021) demonstrated it outside politics. Two very large surveys, one with 250,000 responses per week, overstated US COVID-19 vaccine uptake by up to 17 percentage points against the administrative record. A much smaller random-sample survey was close. Size was not merely unhelpful. It made the confidence intervals narrow enough to exclude the truth.

The AAPOR reviews are the empirical case. Kennedy et al. (2018) found that 2016 state polls over-represented college graduates and largely failed to weight on education, which is the weighting failure above exactly. The 2020 task force found error in the same direction again, and was unable to identify the mechanism. That is the uncomfortable part.

Groves (2006) supplies the caution that keeps this from becoming a slogan. Nonresponse bias is not a function of the non-response *rate*; it depends on the covariance between responding and the measured variable. A 1% response rate is not automatically fatal, and a 60% response rate is not automatically safe. This is why "response rates have collapsed" is a reason for concern rather than a finding.

Bailey (2024) argues the discipline should stop treating non-response as a nuisance to be weighted away and start modeling it explicitly, with assumptions stated and tested. His central claim is close to the one the gap table makes: the quantity that should be published alongside a poll is its assumed response model, and nobody publishes one.

The counter-case deserves a hearing. Polls remain excellent at large differences and poor at small ones. A poll showing a twenty-point race is almost certainly right about who leads. National polling error in 2024 was not historically unusual. The honest position is that the direction of error has been consistent across three cycles without a settled explanation — which is worse than a large error with a known cause.

15 What a built country can testify to

The strength of this evidence is that the answer was known before the poll was taken, which is the one thing no real election ever offers. Every poll above can therefore be graded rather than argued about. The mechanism is visible because it was made small on purpose: one variable, one response gap, one consequence. And the central result is arithmetic rather than luck. The value the biased polls converge on, 53.80%, is algebra rather than a product of the random seed, and it would not move if every draw in this chapter were taken again.

The costs of that clarity are five, and they run in one direction. **It is a simulation**: the response gap was invented, and nothing here is evidence about any real pollster. **Real non-response is not one variable** but many, tangled together and shifting between elections. The weighting grid used a single tidy variable so the mechanism could be seen, and reality offers no such favour.

The gap here is fixed and known, which in practice it is neither. A pollster who knew the gap would already have corrected it. **Sampling here is independent draws**, while real polls cluster, stratify and re-contact, all of which change the effective sample size in ways the square-root rule does not capture.

And **the 2024 state file, though real, is thin**: percentages and electoral votes, no counts, no archive of polls. It shows how hard the task was, rather than how any particular poll performed at it.

Four questions stay open, and the first is open by construction. How large the real response gaps are cannot be measured, because a gap that could be measured would already have been weighted away. Whether the recent polling errors share a cause is suggestive and unexplained: three cycles in the same direction is a pattern, not a mechanism. Whether the industry's proposed fix is even possible depends on assumptions about non-respondents that no data can validate, which is the objection Bailey's own proposal has to answer. And what a poll should publish has no accepted answer at all. There is no standard format for reporting error that does not come from sampling, which is why the only number on the page is the one describing the smallest problem a poll has.

That is the sentence to carry out. The 5 states that decided 2024 were closer than a poll's advertised precision, and the advertised precision is the part of a poll's error that was never the problem.

16 Sources

Data

- `pres2024_states.csv` — 2024 presidential results by state, from the compilation used earlier in the course. Everything else in this document is simulated.

The data itself

Every figure above is drawn from these tables. They are plain CSV, they open in a spreadsheet, and they are yours to take further.

`pres2024_states.csv`

Your turn

None of these is answered above, and every one can be answered from the tables you just opened.

- `pres2024_states.csv` holds the real result in every state, with `margin` and `ev`. Sort by `margin`. How many electoral votes sat inside a margin a normal poll could not resolve?
- Take the states with the smallest margins and add their `ev` together. What is the smallest set of states a polling error would have to get wrong to change the outcome?
- A margin of error covers sampling and nothing else. List three things in this file's states that could go wrong outside the sampling, and say which would be hardest to detect.
- Pick a state and imagine a poll off by three points in the same direction everywhere. Work through `pres2024_states.csv` and see how many states flip.

Academic literature

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